13.12 Congruence and Similarity

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<tr><th colspan="5">Lesson Introduction</th></tr>
<tr><td>Benchmark</td><td colspan="2">MA.912.GR.3.3 Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles, and quadrilaterals.</td><td>Learning Target</td><td>I can use coordinate geometry to justify congruence and similarity in polygons.</td></tr>
<tr><td rowspan="2">Benchmark Coherence</td><td colspan="2">Building On</td><td colspan="2">Working Toward</td></tr>
<tr><td colspan="2">MA.8.GR.2 Understand similarity and congruence using models and transformations.</td><td colspan="2">N/A</td></tr>
<tr><td>Essential Questions</td><td colspan="4">How do we determine if two figures are congruent, similar, or neither?
If we are given only coordinate points, what ways can we determine if two figures are congruent? Similar? Neither?</td></tr>
<tr><td>Lesson Narrative</td><td colspan="4">In this lesson, students explore whether polygons are congruent, similar, or neither by finding the length of each of the sides, given only coordinate points. In the 13.12.3 Guided Practice and 13.12.4 Your Turn!, students apply their understanding of congruent and similar polygons by justifying their choice using definitions, theorems, or properties.</td></tr>
<tr><td rowspan="3">Component Summary</td><td>13.12.1 Warm-Up
(~5 mins)</td><td>Notice and Wonder
(SE p. 348)</td><td>13.12.3 Guided Practice
(10 mins)</td><td>Determining if Polygons are congruent, similar, or neither (SE p. 351)</td></tr>
<tr><td>13.12.2 Exploration
(15 mins)</td><td>Congruence and Similarity in Polygons
(SE p. 349)</td><td>13.12.4 Your Turn
(10 mins)</td><td>Practice determining if polygons are congruent, similar, or neither (SE p. 353)</td></tr>
<tr><td>13.12.5 Wrap-Up
(~5 mins)</td><td colspan="3">Summarizing the Lesson (SE p. 354)</td></tr>
<tr><td rowspan="2">Resources</td><td>Glossary</td><td colspan="2">Materials</td><td>Additional Preparation</td></tr>
<tr><td><u>Key Terms</u>: N/A

<u>Additional Glossary</u>: N/A</td><td colspan="2">Calculators
(Optional) Graph paper / Graphing utility
(Optional)Patty paper</td><td>N/A</td></tr>
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<tr><td rowspan="4">Teacher Tips</td><th>Common Misconceptions</th><th>Things to Consider</th></tr>
<tr><td>Students may have difficulties identifying the difference in congruence and similarity in two figures, which either can be a result in them not understanding their definitions, and/or not being able to distinguish between the two when given two figures.</td><td>Consider having a list of various definitions, theorems, and properties that students have learned throughout the unit and previous units that could be used as justification when working through the 13.12.3 Guided Practice.
As a cumulating end of the course activity, consider having students also utilize their knowledge of transformations by noting the transformations that show congruence or similarity for all figures. Watch for students who miss important details, such as the center of dilation.
If time is an issue, then consider assigning 13.12.4 Your Turn! for homework.</td></tr>
<tr><th>Literacy and Language Routines</th><th>Mathematical Thinking and Reasoning (MTR) Standard</th></tr>
<tr><td>Think-Pair-Share
In the 13.12.1 Warm-Up, students are shown an image and they are asked about what they notice and wonder. Consider opportunities and ways that students can share what they notice and wonder with their peers, such as in Gallery Walks or as a class discussion.</td><td>MA.K12.MTR.5.1: Patterns and Structure
In the 13.12.2 Explorations and 13.12.3 Guided Practice, students are focusing on the relevant details within a problem, while also applying previously learned theorems, properties, and definitions to justify whether two figures are congruent or similar. Students are also connecting the similarities among figures that are similar and congruent, and how that will better help them when solving future problems.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">For students who are struggling in the 13.12.2 Exploration with finding the side lengths given only the coordinate points, consider providing them with graph paper or a graphing tool so that they can see the shape and determine which side lengths they are calculating. Also, consider providing graph paper and patty paper when students are identifying the transformations in the same section.

Additional differentiation opportunities are denoted throughout the lesson guide.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
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13.12.1 Warm-Up: Notice and Wonder

Component Overview: *Students are noticing and wondering about a sculpture seen in Orlando, Florida. Consider opportunities for students to share their ideas with their peers, either in a Gallery Walk, partner share, or class discussion.*

13.12.1 Warm-Up: Notice and Wonder

Brickley, the LEGO sea serpent in Lake Buena Vista at Disney Springs near Orlando, Florida, was made in 1997 from 170,000 LEGO bricks and is 30 feet long.

1. What do you notice?
   Answers will vary. The sculpture has a lot of symmetry in the points on the tail, face and chest of the sculpture.

2. What do you wonder?
   Answers will vary. How often do Legos have to be replaced?

Consider opportunities for students to share what they notice and wonder with their peers. Consider doing a simple partner share where they discuss with their partner. Or, consider a Gallery Walk where students can see multiple classmates' responses, and possibly even write down one thing that they notice and one thing that they wonder from another student.

13.12.2 Exploration: Congruence and Similarity

Component Overview: *In this Exploration, students are exploring if two polygons are congruent, similar, or neither based off of their side lengths, which they are expected to find using the given coordinate points. Consider providing students who are having difficulties visualizing the polygon with graph paper or a graphing tool for them to find the side lengths.*

13.12.2 Exploration: Congruence and Similarity

1. Parallelogram $DKTS$ has vertices at $D(-2,4)$, $K(-4,1)$, $T(-9,3)$, and $S(-7,6)$.

   Complete the table.

Line Segment	Length
$\overline{DK}$	$DK = \sqrt{13} \approx 3.606$
$\overline{KT}$	$KT = \sqrt{29} \approx 5.385$
$\overline{TS}$	$TS = \sqrt{13} \approx 3.606$
$\overline{SD}$	$SD = \sqrt{29} \approx 5.385$

2. Parallelogram $WRCN$ has vertices at $W(0,1)$, $R(2,-2)$, $C(7,0)$, and $N(5,3)$.

   Complete the table.

Line Segment	Length
$\overline{WR}$	$WR = \sqrt{13} \approx 3.606$
$\overline{RC}$	$RC = \sqrt{29} \approx 5.385$
$\overline{CN}$	$CN = \sqrt{13} \approx 3.606$
$\overline{NW}$	$NW = \sqrt{29} \approx 5.385$

3. What do you notice about the lengths of the line segments in parallelograms $DKTS$ and $WRCN$?
   The corresponding side lengths are congruent.

This Exploration is scaffolded to allow students to engage with it with their partner, rather than being guided through by the teacher. The concepts of congruence and similarity are not new to the student. Monitor while students are working to determine if there are any questions that need to be debriefed before moving to 13.12.3 Guided Practice. Consider highlighting student voices to share rich discussions or comments overheard while you circulated, or to address common misconceptions that were evident among the majority of the class. Students may revise their thinking throughout the Exploration as they engage with each new question.

Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole-group Bring It Together.

Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed Exploration.

For students who are struggling with finding the side lengths given only the coordinate points, consider providing them with graph paper or a graphing tool so that they can see the shape and determine which side lengths they are calculating.

4. Complete the statement.

   Parallelogram $DKTS$ may be
   - o congruent
   - o similar

   to parallelogram $WRCN$ because all corresponding sides are
   - o proportional.
   - o congruent.

5. Work with your partner to describe how the parallelograms are congruent or similar using transformations. State which image you used as the pre-image.
   Answers will vary. Parallelogram $DKTS$ is the pre-image. When parallelogram $DKTS$ is reflected across the y-axis, horizontally translated left 2 units, and vertically translated down 3 units, the result is parallelogram $WRCN$.

6. Isosceles trapezoid $MPHN$ has vertices at $M(2,-6)$, $P(3,-3)$, $H(5,-2)$, and $N(8,-3)$.

   Complete the table.

Line Segment	Length
$\overline{MP}$	$\sqrt{10} \approx 3.162$
$\overline{PH}$	$\sqrt{5} \approx 2.236$
$\overline{HN}$	$\sqrt{10} \approx 3.162$
$\overline{NM}$	$\sqrt{45} \approx 6.708$

- *If parallelogram DKTS was similar to parallelogram WRCN, what would the side lengths look like?*
- *How would the side lengths change?*

13.12.2 Exploration: Congruence and Similarity (continued)

7. Isosceles trapezoid $WVYZ$ has vertices at $W(4,12)$, $V(6,6)$, $Y(10,4)$, and $Z(16,6)$.
Complete the table.

Line Segment	Length
$\overline{WV}$	$\sqrt{40} \approx 6.325$
$\overline{VY}$	$\sqrt{20} \approx 4.472$
$\overline{YZ}$	$\sqrt{40} \approx 6.325$
$\overline{ZW}$	$\sqrt{180} \approx 13.416$

8. What do you notice about the lengths of the line segments in isosceles trapezoids $MPHN$ and $WVYZ$?
The line segments in isosceles trapezoid $WVYZ$ are the side lengths of the isosceles trapezoid $MPHN$ multiplied by 2.

9. Complete the statement.

Isosceles trapezoid $MPHN$ is
- o congruent
- o similar

to isosceles trapezoid $WVYZ$ because all corresponding sides are
- o proportional.
- o congruent.

10. Work with your partner to describe how the trapezoids are congruent or similar using transformations. State which image you used as the pre-image.
Answers will vary. Isosceles trapezoid $MPHN$ has been reflected across the x-axis and then dilated by a scale factor of 2 about the origin to create isosceles trapezoid $WVYZ$.

For students struggling with finding the transformations, provide students with graph paper and patty paper so that they can visually see the correct sequence of transformations.

For students that are struggling with identifying transformations, consider providing students with graph paper and patty paper.

- *How can we determine if polygons are similar or congruent?*
- *What characteristics do shapes that are congruent have?*
- *What characteristics do shapes that are similar have?*

13.12.3 Guided Practice: Congruence and Similarity

Component Overview: *Students are continuing their understanding of similarity and congruence from the previous section by determining if two polygons are congruent and similar, but instead of only calculating the side length, they are also asked to justify their choice by using definitions, properties, and theorems that they have learned.*

13.12.3 Guided Practice: Congruence and Similarity

1. Right triangle CKR has vertices at $C(-2,5)$, $K(-8,5)$, and $R(-8,2)$. Right triangle DPM has vertices at $D(-2,-4)$, $P(4,-4)$, and $M(4,-1)$.

   a. Determine if ΔCKR and ΔDPM are congruent, similar, or neither.
   The triangles are congruent because the side lengths are equal.

   b. What definition, theorem, or property justifies your conclusions?
   Answers will vary. The SSS congruence theorem.

   c. Describe the transformations that would show that the figures are congruent or that they are similar. State which image you used as the pre-image.
   Answers will vary. Let triangle CKR be the pre-image. When you rotate ΔCKR $180°$ about point C and then translate down 9 units the result is triangle DPM.

   d. Ask a classmate for their description. Record their description and write your summary of their description. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Description	My Summary of Their Description	Initials
Answers will vary.		

Allow informal observation throughout 13.12.2 Exploration to dictate how "guided" this Guided Practice should be for you and your classroom. This could mean small group instruction for some while others attempt independently, or it could mean teacher facilitation of practice with students actively working as a whole group.

Consider having an anchor chart with multiple definitions, theorems, and properties that were learned in this unit and in earlier units so that students could reference when justifying whether two polygons are congruent or similar.

What is another justification, different than your original one, that also supports why triangle CKR and DPM are congruent?

For students who need a challenge, consider having them find another series of transformations in question 1c that would also show that the two figures are congruent.

13.12.3 Guided Practice: Congruence and Similarity (continued)

2. Parallelogram $HGNC$ has vertices at $H(5,-1)$, $G(2,3)$, $N(9,3)$, and $C(12,-1)$. Parallelogram $VYZW$ has vertices at $V(-1,1.5)$, $Y(0.5,4)$, $Z(-3,4)$, and $W(-4.5,1.5)$.

   a. Shin and Kendal are determining if parallelograms $HGNC$ and $VYZW$ are congruent, similar, or neither. Their work is shown.

Shin's Work	Kendal's Work
Parallelograms $HGNC$ and $VYZW$ are similar because $\frac{ZY}{NG} = \frac{VW}{HC}$	Parallelograms $HGNC$ and $VYZW$ are neither because $\frac{ZY}{NG} \neq \frac{YV}{GH}$

   b. Whose work do you agree with?
   Kendal's work.

   c. Write a note to the person who is incorrect explaining their mistake.

Answers will vary.

Two pairs of sides being proportional does not prove that two parallelograms are similar. More information about the other sides and the angles formed is needed.

Means
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Start
To
Acquire
Knowledge
Experience
Skills

Consider spending time reviewing the transformations that students wrote. Help students recall information that they may have forgotten. Including transformations in this lesson was purposely done so that students are using a spaced recall, which research shows helps students remember concepts. This could help students with end-of-year summative assessments.

For students who are struggling with determining whose work they agree with, consider having students first find the side lengths and/or graph the shapes before they make their decision.

Consider having students share their notes. Encourage students to be thorough in their explanations.

13.12.4 Your Turn!

Component Overview: *Students continue their understanding that was bult in the last two sections by determining if two figures are congruent, similar, or neither, given only their coordinate points.*

13.12.4 Your Turn!

Complete the table by determining if the given polygons are congruent, similar, or neither.

1. ΔSTK: $S(-6,8)$, $T(-10,5)$, and $K(-6,1)$
   ΔNRC: $N(3,7.5)$, $R(-3,3)$, and $C(3,-3)$
   - Congruent
   - Similar
   - Neither

2. Trapezoid $MPWY$: $M(3,4)$, $P(7,6)$, $W(10,5)$, and $Y(4,2)$
   Trapezoid $LKGH$: $L(-2,6)$, $K(-8,9)$, $G(-14,7)$, and $H(-4,2)$
   - Congruent
   - Similar
   - Neither

3. Kite $XZWY$: $X(-7,6)$, $Z(-6,9)$, $W(-3,10)$, and $Y(-1,4)$
   Kite $FGDK$: $F(2,3)$, $G(1,6)$, $D(-2,7)$, and $K(-4,1)$
   - Congruent
   - Similar
   - Neither

For students who are having difficulties visualizing the figures, consider providing graph paper or a graphing tool so that they can determine if the two figures are congruent, similar, or neither.

What are the ways to determine if two figures are congruent? Similar? Neither?

For students who finish early, consider having them do the following:

- *Create a set of coordinate points of a different figure in question 1 that is congruent to the given figure.*
- *Create a set of coordinate points of a different figure in question 2 that is similiar to the given figure.*
- *Create a set of coordinate points of a different figure in question 3 that is similar to the given figure.*

Also, consider having students create their own figure, and then create 3 different sets of coordinate points that are congruent, similar, and neither to their created figure.

13.12.5 Wrap-Up: Cool Down

Component Overview: *Students write a note to a friend, summarizing the ideas and concepts that they took away from this lesson.*

13.12.5 Wrap-Up: Cool Down

1. Your friend is absent today and will need the notes from class. Write a quick note to teach them how to use coordinate geometry to justify congruence and similarity in polygons.

Answers will vary.

Consider what responses and topics you are looking for students to have in their note that you would consider a student to have understood this lesson. Consider creating a small rubric that would help you when reading over the "notes," and what you would be looking for with a student who completely understood the lesson, versus a student who may need help.